A Contrast-Sensitivity-Constrained Adversarial Patch for Semantic Segmentation

Formal procedure and provenance

Abstract

We define an adversarial patch whose deviation from the image content it covers is bounded in the frequency domain by a model of human contrast sensitivity. The construction rests on an asymmetry: a segmentation network weights the spectrum up to Nyquist uniformly, whereas the human visual system does not, so a perturbation may be admissible to one and not the other. Whether that gap is exploitable for a given architecture is an empirical question, not an assumption of the method. Every step below is stated formally and attributed; five elements have no external source and are marked as such throughout.

1 Notation

$\mathbf{x} \in \mathbb{R}^{3\times H\times W}$	input image, ImageNet-normalised
$\mathbf{x}^{01}$	the same image in $[0,1]$ RGB, $\mathrm{clip}(\mathbf{x} \odot \boldsymbol{\sigma}_n + \boldsymbol{\mu}_n, 0, 1)$
$\mathbf{y}$	ground-truth trainIds in $\{0,\dots,C-1\} \cup \{255\}$; 255 is void
f	frozen segmentation network, $f(\mathbf{x}) \in \mathbb{R}^{C\times H'\times W'}$
$\mathcal{U}$	bilinear upsampling to $H \times W$
$s,\ p = \lfloor Hs \rfloor,\ S$	patch scale, rendered side in pixels, parameter resolution
$\mathbf{z} \in \mathbb{R}^{3\times S\times S}$	the optimised parameter
$\mathcal{F}, \mathcal{F}^{-1}$	real 2-D DFT and inverse, forward-normalised
$\mathcal{K}$	half-spectrum bin index set
$\mathcal{M}$	set of pixels actually composited (silhouette; all of them if square)
τ	perceptual budget, in nominal JND (Sec. 11)

2 Observer model

Contrast sensitivity is defined over cycles per degree of visual angle, whereas an image spectrum is indexed in cycles per pixel. With physical pixel size ρ and viewing distance d, one pixel subtends

$$\theta_{\mathrm{pix}} \;=\; 2\arctan\!\left(\frac{\rho}{2d}\right)\cdot\frac{180}{\pi} \quad \text{degrees,} \tag{1}$$

so an image frequency ν in cycles per pixel corresponds to $u = \nu/\theta_{\mathrm{pix}}$ cycles per degree (Galinska et al., 2026, App. D). We adopt the Barten model (Barten, 2003),

$$\mathrm{CSF}(u) = \frac{M_{\mathrm{opt}}(u)}{k}\left[\frac{2}{T}\left(\frac{1}{X_0^2}+\frac{1}{X_{\max}^2}+\frac{u^2}{N_{\max}^2}\right)\left(\frac{1}{\eta\,\mathrm{p}\,E}+\frac{\Phi_0}{1-\exp(-(u/u_0)^2)}\right)\right]^{-1/2}, \tag{2}$$

$$M_{\mathrm{opt}}(u) = \exp(-2\pi^2\varsigma^2u^2), \tag{3}$$

with the parameterisation of Galinska et al. (2026, App. A): $\varsigma = 0.5/60$, $k = 3$, $T = 0.1$, $X_0 = 60$, $X_{\max} = 12$, $N_{\max} = 15$, $\eta = 0.03$, $\mathrm{p} = 1.2\times10^6$, $E = 500$, $\Phi_0 = 3\times10^{-8}$, $u_0 = 7$. Note

CSF(0) = 0: a uniform field carries no contrast. Where orientation dependence is wanted we substitute the Standard Spatial Observer with its oblique effect (Watson and Ahumada, 2005), under which diagonal carriers are less visible than axis-aligned ones at equal radial frequency.

Scope of the observer assumption. Every visibility statement below is conditional on (ρ, d) through Eq. (1). This is not incidental: Galinska et al. (2026, Sec. 6) identify the assumption of a typical screen and viewing distance as a *conceptual limitation* of any CSF-derived weighting, requiring calibration if optimal values are wanted. We therefore treat (ρ, d) as a reported experimental parameter and sweep it rather than fixing it once.

Why a pixel-space formulation is required. The correspondence between DFT bins and retinal spatial frequency holds only when the perturbation lives in pixel space. Galinska et al. (2026, Sec. 5.1) adopt the same restriction for the same reason: in latent-space models the coordinates of the perturbed variable need not correspond to image spatial frequencies at all. The patch here is optimised directly in $[0, 1]$ RGB, so Eq. (1) is well defined throughout.

3 Per-frequency amplitude budget [no external source]

A grating of Michelson contrast c at frequency u is at detection threshold when $c\,\mathrm{CSF}(u) = 1$ (Barten, 2003). Inverting this into an *amplitude allowance* on the DFT is what turns a perception model into an attack constraint. It does not appear in Galinska et al. (2026), who use CSF as a *weight* over generative timesteps; the function and its constants come from that work, the inversion does not.

On the half-spectrum grid with radial frequency $\nu(\mathbf{k}) = \sqrt{\nu_y^2 + \nu_x^2}$ in cycles per pixel,

$$B(\mathbf{k}) \;=\; \min\left(\frac{\tau}{\kappa\,\mathrm{CSF}\big(\nu(\mathbf{k})/\theta_{\mathrm{pix}}\big)},\; A_{\max}\right) \cdot \mathbb{1}\big[\nu(\mathbf{k}) \geq \nu_{\min}\big]. \tag{4}$$

Contrast conversion $\kappa = 4$. A cosine of peak amplitude A on mean luminance $\bar{L}$ has Michelson contrast $A/\bar{L}$ (Michelson, 1927); under the forward-normalised DFT that component has magnitude $A/2$. Taking $\bar{L} = \frac{1}{2}$ for a $[0, 1]$ image gives $c = 4\,|\mathcal{F}(\cdot)(\mathbf{k})|$. The mean is assumed rather than measured locally, which *under*-estimates visibility on dark content; the discrepancy is absorbed into the calibration of τ.

Low-frequency cutoff $\nu_{\min} = m_c/S$. Since CSF $\to 0$ at DC, Eq. (4) would otherwise grant the near-DC bins the *largest* allowance of any frequency. The Barten rolloff is measured for full-field gratings, where the eye adapts to a slow luminance ramp; inside a patch of side S a component below $1/S$ cycles per pixel completes no cycle and is a brightness offset against the surround, which the patch border renders as a visible edge. We require $m_c = 2$ cycles across the patch. Omitting this inverts the intended spectrum: measured radial power then concentrates in the lowest bin rather than the highest.

4 Spectral reparameterisation [no external source]

Rather than projecting onto the feasible set, we reparameterise so the constraint holds by construction:

$$Z = \mathcal{F}(\mathbf{z}), \qquad \hat{Z}(\mathbf{k}) = B(\mathbf{k})\,\frac{Z(\mathbf{k})}{\sqrt{1 + |Z(\mathbf{k})|^2}}, \qquad \boldsymbol{\delta}_0 = \mathcal{F}^{-1}(\hat{Z}), \tag{5}$$

so $|\hat{Z}(\mathbf{k})| < B(\mathbf{k})$ for every $\mathbf{k}$ and any $\mathbf{z}$. The scaling acts on the complex coefficient, preserving phase exactly and avoiding differentiation of $\arg Z$ at the origin. A hard projection would zero the gradient of every component resting on its bound—precisely where an adversarial component settles—whereas Eq. (5) is smooth everywhere and attains the bound only in the limit.

5 Visibility normalisation

Detection of a compound stimulus is not decided by its most visible component: the visual system pools evidence across spatial-frequency and orientation channels, standardly modelled as a Minkowski norm with exponent $\beta \approx 3$–4 (Quick, 1974; Robson and Graham, 1981; Watson, 1979). With $\mathcal{M}$ the composited support,

$$V_\beta(\boldsymbol{\delta}) = \left[\sum_{\mathbf{k}\in\mathcal{K}} \Big(\kappa \, |\mathcal{F}(\boldsymbol{\delta} \odot \mathcal{M})(\mathbf{k})| \; \mathrm{CSF}\big(\nu(\mathbf{k})/\theta_{\mathrm{pix}}\big) \Big)^{\beta} \right]^{1/\beta}, \tag{6}$$

and the residual is rescaled to exactly the requested budget,

$$\boldsymbol{\delta}_1 = \frac{\tau}{V_\beta(\boldsymbol{\delta}_0)} \, \boldsymbol{\delta}_0. \tag{7}$$

Shape and scale are deliberately separate: Eq. (5) bounds each bin but says nothing about their sum, while Eq. (7) alone would place energy wherever the attack prefers. Equality rather than inequality is used because the constraint is active at every step and equality is differentiable where $\min(\cdot)$ is not.

Taking the maximum over $\mathbf{k}$ instead of the Minkowski sum (the $\beta \to \infty$ limit) is the more permissive criterion and should not be used: a residual scoring 0.235 under it was measured reaching ± 7.65 in pixel space.

6 Dynamic-range fit [NO EXTERNAL SOURCE]

The composite must lie in $[0, 1]$, but clipping is a hard nonlinearity and generates broadband harmonics at exactly the frequencies where CSF peaks, destroying the guarantee of Eqs. (4)–(7): a residual built to 1.0 nominal JND was measured at 5.0 after clipping, with only 8% of pixels affected. A *uniform* rescale is the only operation leaving the spectral shape invariant, since it scales every Fourier coefficient equally and hence $V_\beta(c\boldsymbol{\delta}) = c\, V_\beta(\boldsymbol{\delta})$. With per-pixel headroom

$$a(u,v) = \begin{cases} 1 - \mathbf{r}(u,v), & \delta_1(u,v) > 0, \\ \mathbf{r}(u,v), & \text{otherwise}, \end{cases} \qquad c^\star = \mathrm{clip}\bigg(Q_q \Big\{ \frac{a(u,v)}{|\delta_1(u,v)|} \; : \; (u,v) \in \mathcal{M} \Big\}, 0, 1 \bigg), \tag{8}$$

we set $\boldsymbol{\delta} = c^\star \boldsymbol{\delta}_1$, where Q_q is the q-quantile ($q = 10^{-3}$). The quantile rather than the minimum admits a negligible fraction of clipped pixels, since a single saturated pixel would otherwise force $c^\star = 0$. Restricting the quantile to $\mathcal{M}$ is essential: pixels outside the composited region cannot affect the result yet would otherwise determine its scale. With a cutout reference padded in transparent black, omitting that restriction drives $c^\star$ to zero and the attack becomes the unmodified reference.

7 Reference and patch

The base is the image content the patch replaces, at the resolved placement (t, ℓ):

$$\mathbf{r} = \mathrm{Resize}_{S\times S}\Big(\mathbf{x}^{01} [\, t : t{+}p, \; \ell : \ell{+}p \,] \Big), \qquad \mathbf{p} = \mathrm{clip}(\mathbf{r} + \boldsymbol{\delta}, 0, 1). \tag{9}$$

$\mathbf{r}$ is treated as a constant, $\nabla_{\mathbf{r}} \equiv 0$. Because $\mathbf{r}$ equals the covered content, replacement by $\mathbf{p}$ is identical to *adding* $\boldsymbol{\delta}$, so $\tau \to 0$ recovers the unperturbed image exactly and the no-attack baseline is a no-op by construction rather than merely a weak attack.

8 Application

Following the localised-patch threat model of Brown et al. (2017), with optional silhouette mask after Tan et al. (2021),

$$\tilde{\mathbf{p}} = \left(\mathrm{Resize}_{p\times p}(\mathbf{p}) - \boldsymbol{\mu}_n\right) \oslash \boldsymbol{\sigma}_n, \tag{10}$$

$$\mathbf{m} = \mathrm{Pad}_{t,\ell}(\nVdash_{p\times p} \odot \mathcal{M}), \tag{11}$$

$$\mathbf{x}^{\mathrm{adv}} = \mathbf{x} \odot (1 - \mathbf{m}) \; + \; \mathrm{Pad}_{t,\ell}(\tilde{\mathbf{p}}) \odot \mathbf{m}. \tag{12}$$

9 Objective

The scored support excludes both void labels and the patch footprint,

$$\Omega = \{(u,v) : \mathbf{y}(u,v) \neq 255\} \;\setminus\; \mathrm{supp}(\mathbf{m}), \tag{13}$$

the latter to prevent the degenerate solution in which the patch itself adopts the attacked appearance rather than influencing its surroundings (Mirsky, 2023). We use CosPGD (Agnihotri et al., 2024),

$$\mathcal{L} = -\frac{1}{|\Omega|} \sum_{(u,v)\in\Omega} \Big[\cos\left(\mathrm{softmax}\,\mathcal{U}f(\mathbf{x}^{\mathrm{adv}})_{:,u,v},\, \mathbf{e}_{\mathbf{y}(u,v)}\right)\Big]_{\mathrm{sg}} \cdot\, \mathrm{CE}(\mathcal{U}f(\mathbf{x}^{\mathrm{adv}})_{:,u,v},\, \mathbf{y}(u,v)). \tag{14}$$

Declared deviations. Agnihotri et al. (2024) do not detach the cosine; $[\cdot]_{\mathrm{sg}}$ removes gradient flow through the weight. They further use sign-SGD with ε-projection, as a PGD variant (Madry et al., 2018); a patch is not an ε-ball perturbation, so Eq. (14) is optimised with Adam (Kingma and Ba, 2015) on the raw gradient.

10 Optimisation and evaluation

$$\mathbf{z}^{(0)} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \qquad \mathbf{z}^{(j+1)} = \mathrm{Adam}(\mathbf{z}^{(j)}, \nabla_{\mathbf{z}}\mathcal{L}), \qquad j = 0, \ldots, K-1. \tag{15}$$

The initialisation is Gaussian rather than zero because Eq. (7) is undefined at $\boldsymbol{\delta}_0 = \mathbf{0}$; Eqs. (5)–(7) are scale-invariant in $\mathbf{z}$, so only the *shape* of the initialisation matters. The network f is frozen throughout and $\nabla_\phi f$ is never formed. The attack is reported as

$$\Delta\mathrm{mIoU}_{\mathrm{remote}} = \mathrm{mIoU}(\mathcal{U}f(\mathbf{x}), \mathbf{y}; \Omega) - \mathrm{mIoU}(\mathcal{U}f(\mathbf{x}^{\mathrm{adv}}), \mathbf{y}; \Omega), \tag{16}$$

the mean taken over classes *present in* $\mathbf{y}|_\Omega$ so that both terms share a denominator (Cordts et al., 2016). Counting a class merely because it is predicted makes the two means run over different class sets, and a rare class entering or leaving the denominator shifts ΔmIoU by roughly $\mathrm{mIoU}/(n-1)$ independently of any pixel-level change. Spectral placement of the learned residual is reported by radially averaged power spectral density (Ruzanski and Chandrasekar, 2011), and perceptual similarity to $\mathbf{r}$ by LPIPS (Zhang et al., 2018), computed after training and never entering Eq. (14).

11 The budget parameter τ

Definition. τ is the permitted Minkowski visibility of the residual, in units of the Barten detection threshold under the assumed geometry: by Eq. (7) the composited perturbation satisfies $V_\beta(\boldsymbol{\delta}_1) = \tau$ exactly, so $\tau = 1$ nominally places it *at* threshold and $\tau < 1$ below. It plays the role α plays in Tan et al. (2021)—the single knob trading attack strength against perceptibility—and the curve over τ, not any single operating point, is the deliverable.

τ acts once, not twice. Although τ appears in both Eq. (4) and Eq. (7), its occurrence in the budget is inert whenever $A_{\max}$ is inactive: B is then proportional to τ, hence so are $\boldsymbol{\delta}_0$ and $V_\beta(\boldsymbol{\delta}_0)$, and the ratio in Eq. (7) cancels it. Measured at $S = 128$ and the default geometry, varying τ in Eq. (4) alone over $[0.05, 1]$ leaves the residual bit-identical, and $A_{\max} = 0.25$ binds on no bin until $\tau \gtrsim 4$. Equation (4) therefore sets the residual's *shape*; Eq. (7) sets its *scale*.

Operating range. Below the knee of Eq. (8) realised visibility tracks τ exactly and the residual scales linearly:

τ	0.05	0.10	0.25	0.50
residual RMS	0.0049	0.0098	0.0245	0.0490

Above the knee the range fit binds, attained visibility saturates and τ ceases to control anything. The knee is reference-dependent, being set by the headroom of $\mathbf{r}$, and lay near $\tau \approx 0.4$ for the references used here. Values above it must not be reported as though achieved.

Why τ is *nominal.* Three approximations stand between τ and a psychophysical JND, each biasing it: the mean luminance in κ is assumed rather than measured; the Minkowski sum in Eq. (6) runs over DFT bins rather than perceptual channels; and Eq. (1) fixes a viewing geometry that Galinska et al. (2026, Sec. 6) identify as needing calibration. τ should therefore be calibrated against rendered patches and reported as a relative axis. A claim of imperceptibility requires a study with human observers, which this procedure does not constitute.

Not the interpolation parameter of Galinska et al. (2026). Their α in $w_{\text{final}} = \alpha w_{\text{CSFlow}} + (1-\alpha) w_{\text{base}}$ blends a perception-driven weighting with a baseline one over generative timesteps. It is not a perceptual budget and bounds nothing; the two share no role beyond both being scalars derived from the same CSF.

Bounded by CSF **does not imply statistically inconspicuous.** Natural image spectra follow an approximate power law, so energy placed near Nyquist is atypical of the data precisely where the eye is least sensitive (Falck et al., 2025; Dieleman, 2024). Invisibility to a human observer and indistinguishability from natural image statistics are distinct properties; only the former is constrained here, and the RAPSD of Sec. 10 is what makes the latter measurable.

12 Provenance

Sec.	Element	Source
2	cycles/pixel $\rightarrow$ cycles/degree	Galinska et al. (2026, App. D)
2	Barten CSF; parameter values	Barten (2003); params Galinska et al. (2026, App. A)
2	Standard Spatial Observer, oblique effect	Watson and Ahumada (2005)
2	pixel-space requirement; geometry caveat	Galinska et al. (2026, Sec. 5.1, 6)
3	*inversion of CSF into a budget*	**no external source**
3	Michelson contrast, $\kappa = 4$	Michelson (1927)
3	*low-frequency cutoff for a patch*	**no external source**
4	*smooth spectral reparameterisation*	**no external source**
5	Minkowski / probability summation	Quick (1974); Robson and Graham (1981); Watson (1979)
6	*uniform-rescale range fit*	**no external source**
7	*covered content as residual base*	**no external source**
8	localised patch application	Brown et al. (2017)
8	silhouette masking	Tan et al. (2021)
9	footprint exclusion	Mirsky (2023)
9	CosPGD objective	Agnihotri et al. (2024)
9	PGD lineage; declared deviation	Madry et al. (2018)
10	Adam	Kingma and Ba (2015)
10	mIoU protocol, Cityscapes	Cordts et al. (2016)
10	RAPSD	Ruzanski and Chandrasekar (2011)
10	LPIPS, evaluation only	Zhang et al. (2018)
11	natural-image spectral statistics	Falck et al. (2025); Dieleman (2024)

Five elements carry no external source and are contributions of this work: the budget inversion (Eq. 4), the patch-specific low-frequency cutoff, the smooth spectral reparameterisation (Eq. 5), the uniform-rescale range fit (Eq. 8), and the use of the covered content as the residual base (Eq. 9). From Galinska et al. (2026) we take a function, its constants, a unit conversion and two scoping caveats—not a method: their contribution is a weighting over flow-matching timesteps, which has no analogue here.

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